

Alice 3 TOF Module and Stave Layout oTOF Update (92cm)

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science & innovation
Department:
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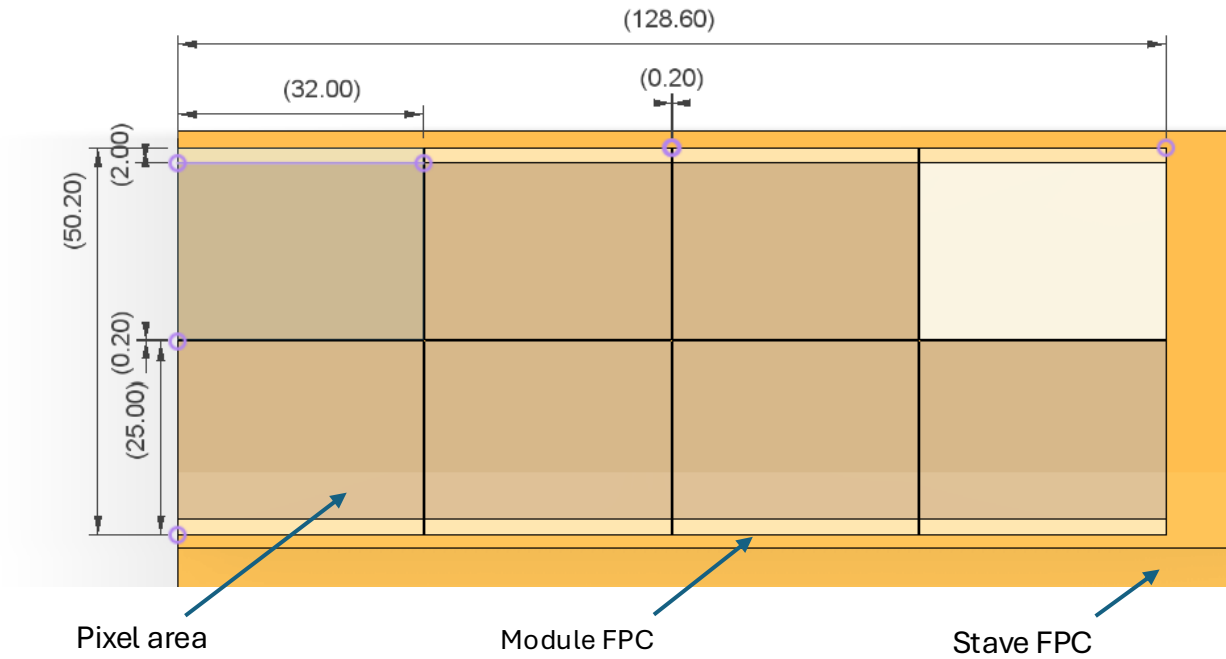


Module Layout and Design – Remains the Same

Sensor Dimension	Value	Unit
Width	25	mm
Length	32	mm
FEE area	64	mm ²
Active pixel area	736	mm ²
Active pixel width	23	mm
Active pixel length	32	mm

Module Dimension	Value	Unit
Sensor spacing	0.2	mm
Module sensor aggregation	8	
Module width	50.2	mm
Module length	128.6	mm
Module spacing	0.2	mm

- Sensors glued to module Flexible Printed Circuit Board (FPC)
- Modules connect to Stave FPC
- Readout electronics on Stave FPC

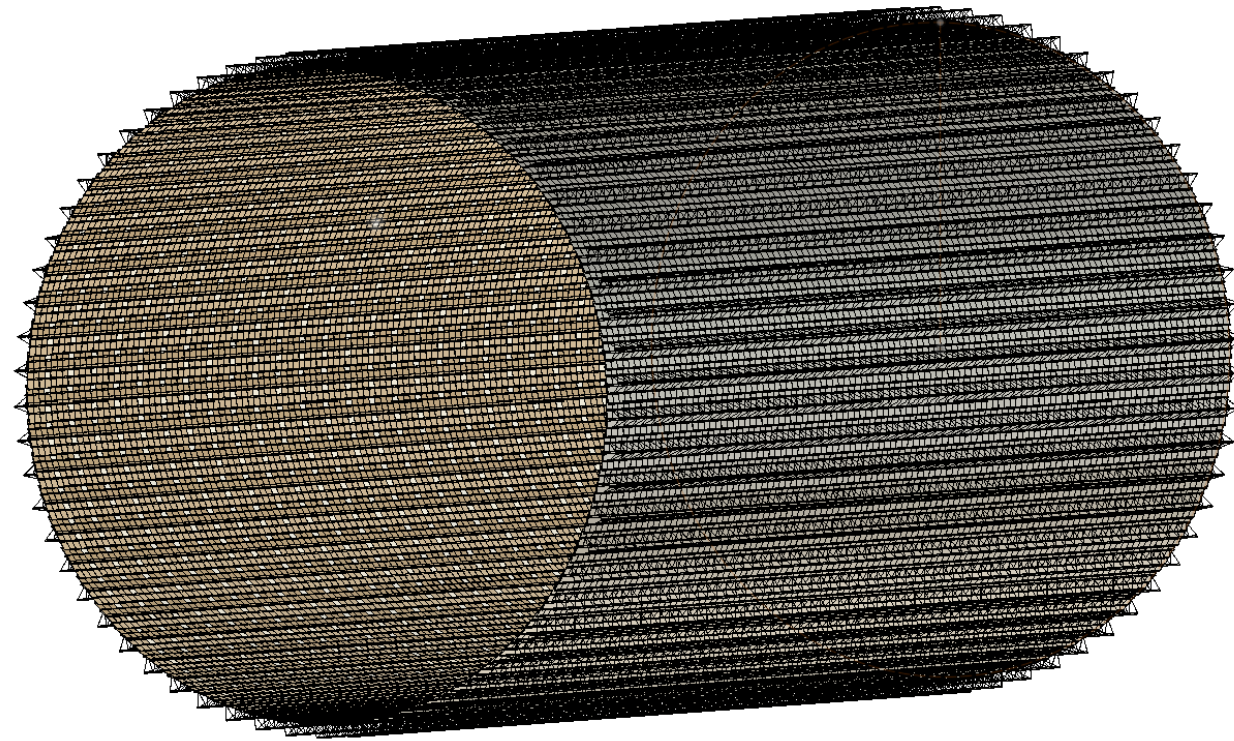
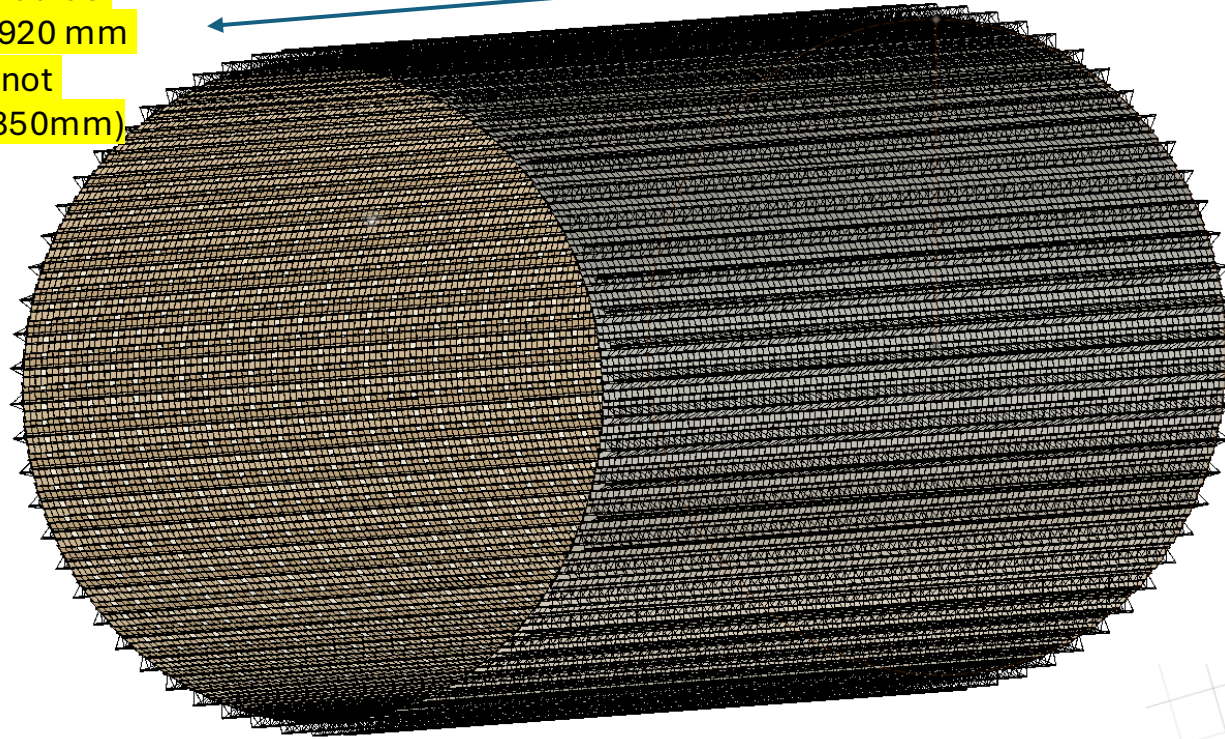


Cooling require for $\sim 200\text{mW/cm}^2$

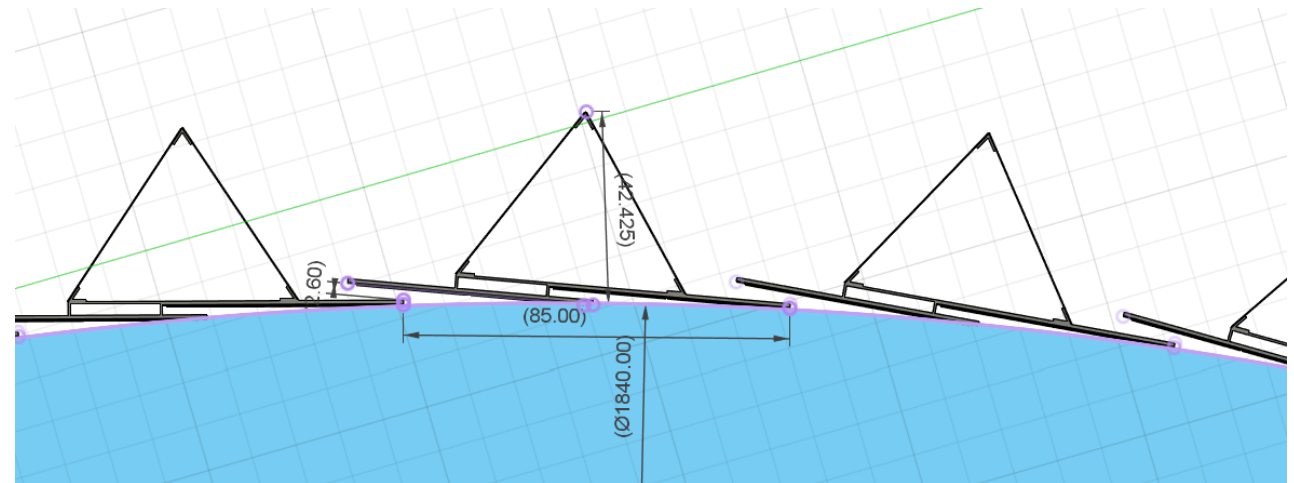
oTOF Stave

Half stave length: 35000 (not 3477 mm)

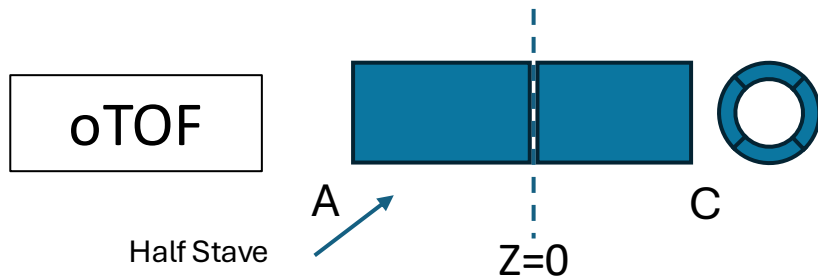
Radius:
920 mm
(not
850mm)



C Side Half Stave Layout



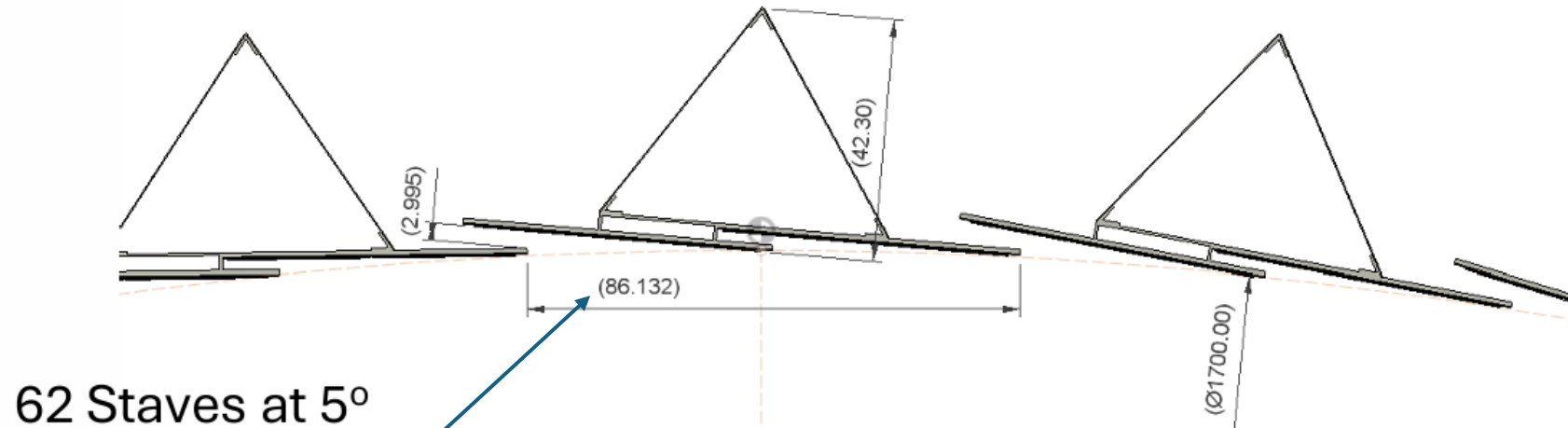
A Side Half Stave Layout



68 Staves at 5°

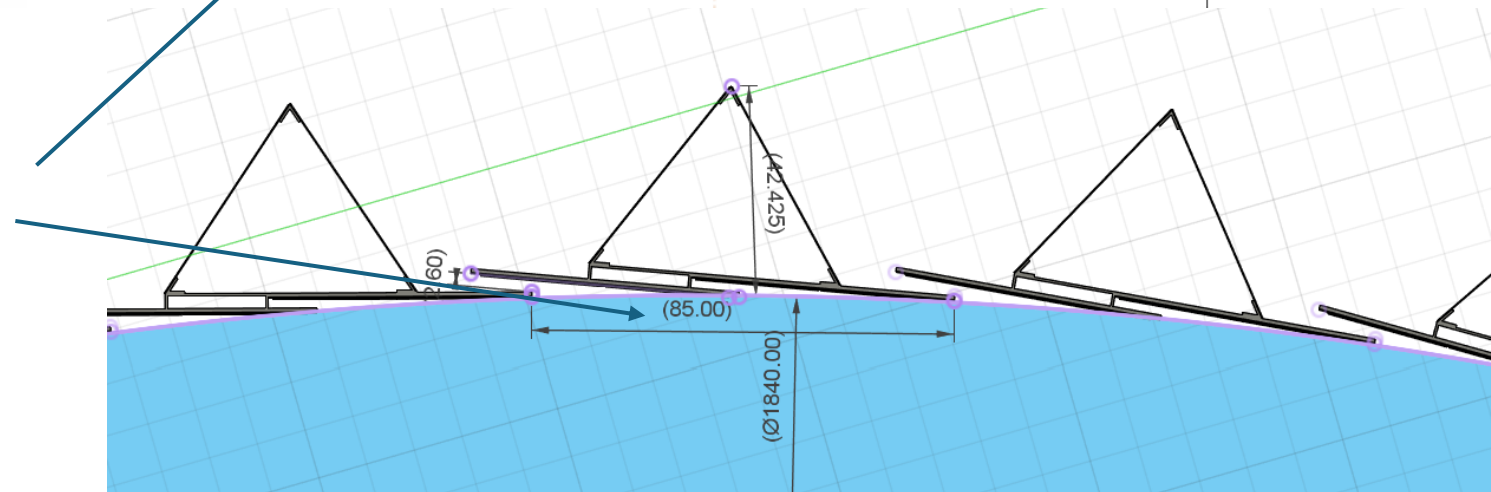
oTOF Update

- V1 R- 85cm



- V1 R- 92cm

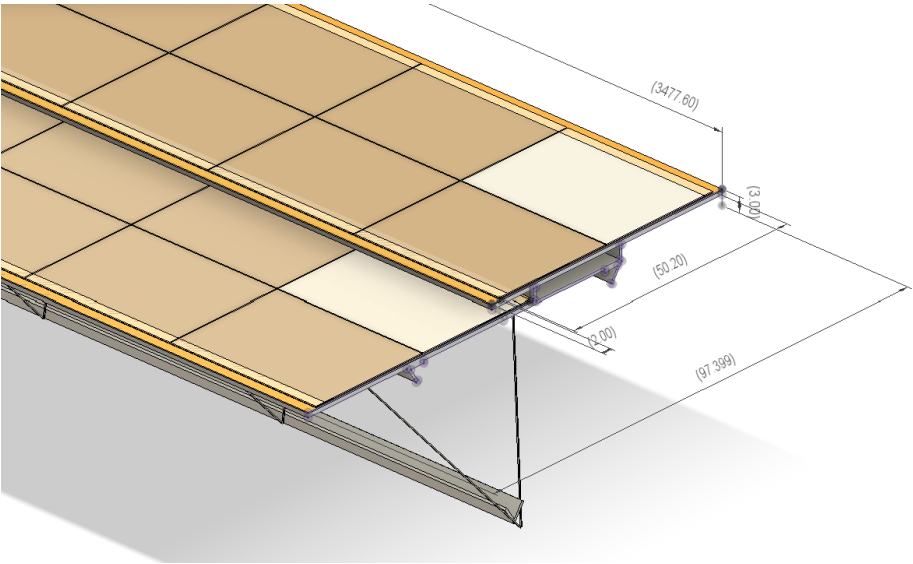
1.13 mm additional overlap



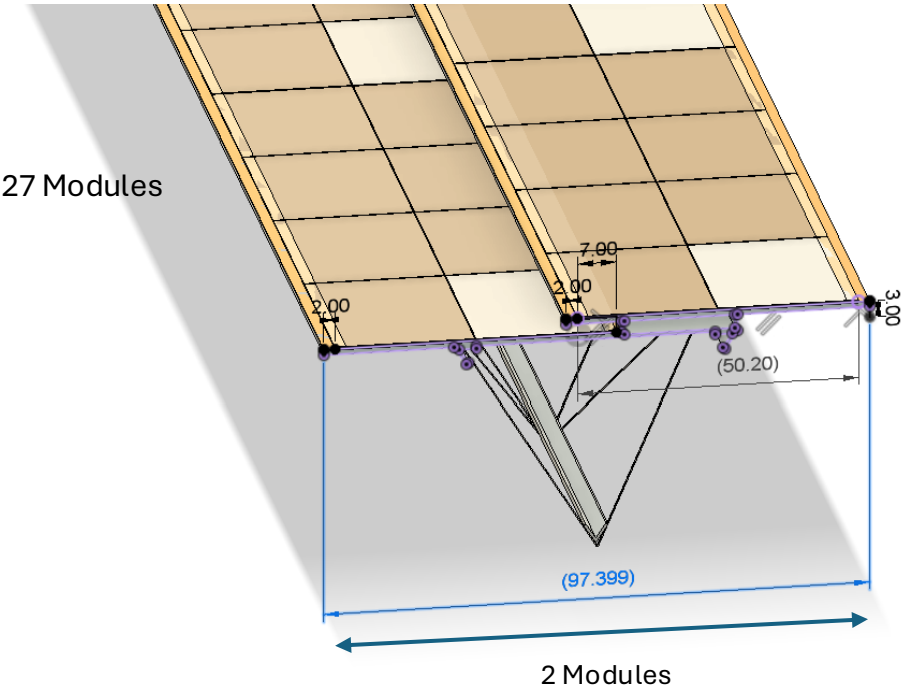
68 Staves at 5°

oTOF Stave

oTOF Dimensions	Value V1 R=85cm	Value V2 R=92 cm	Unit
Radius	850	920	mm
Full length	6955.2	6955.2	mm
Circumference	5340.7	578.05	mm
Half stave length	3477.6	3477.6	mm
Surface area	37.14	40.20	m ²
Number of Half Staves (C and A side)	124	136	



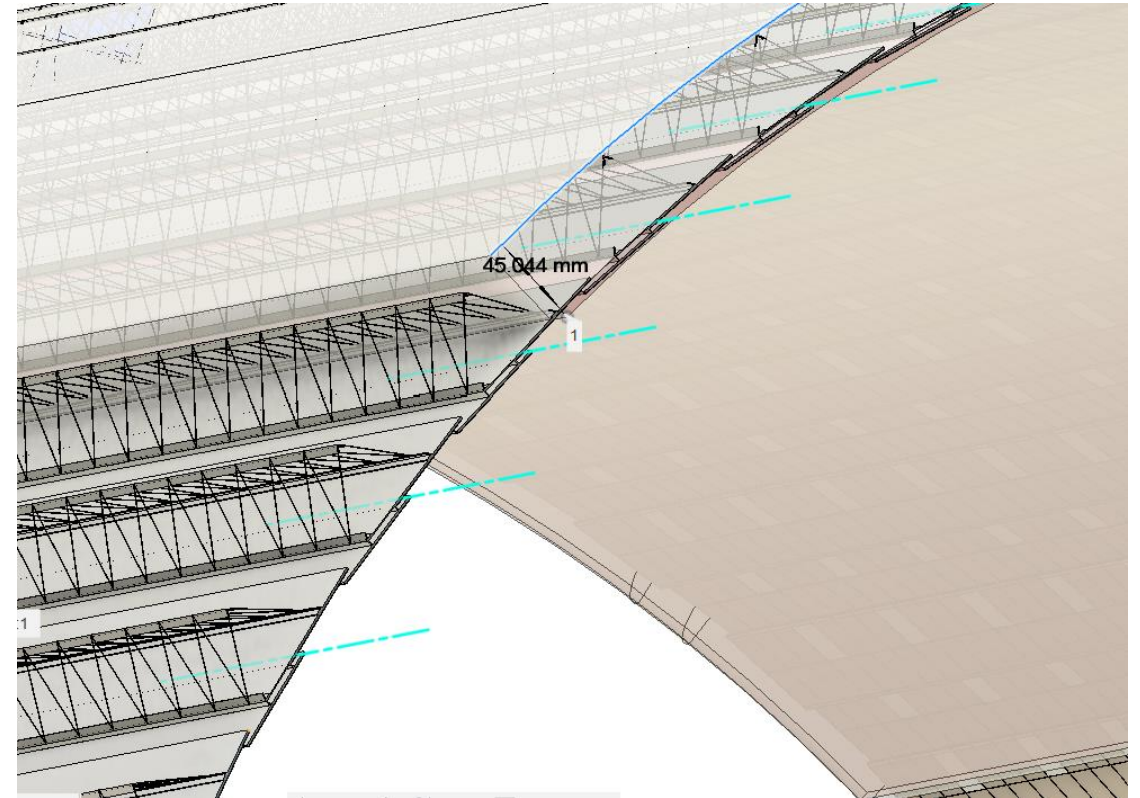
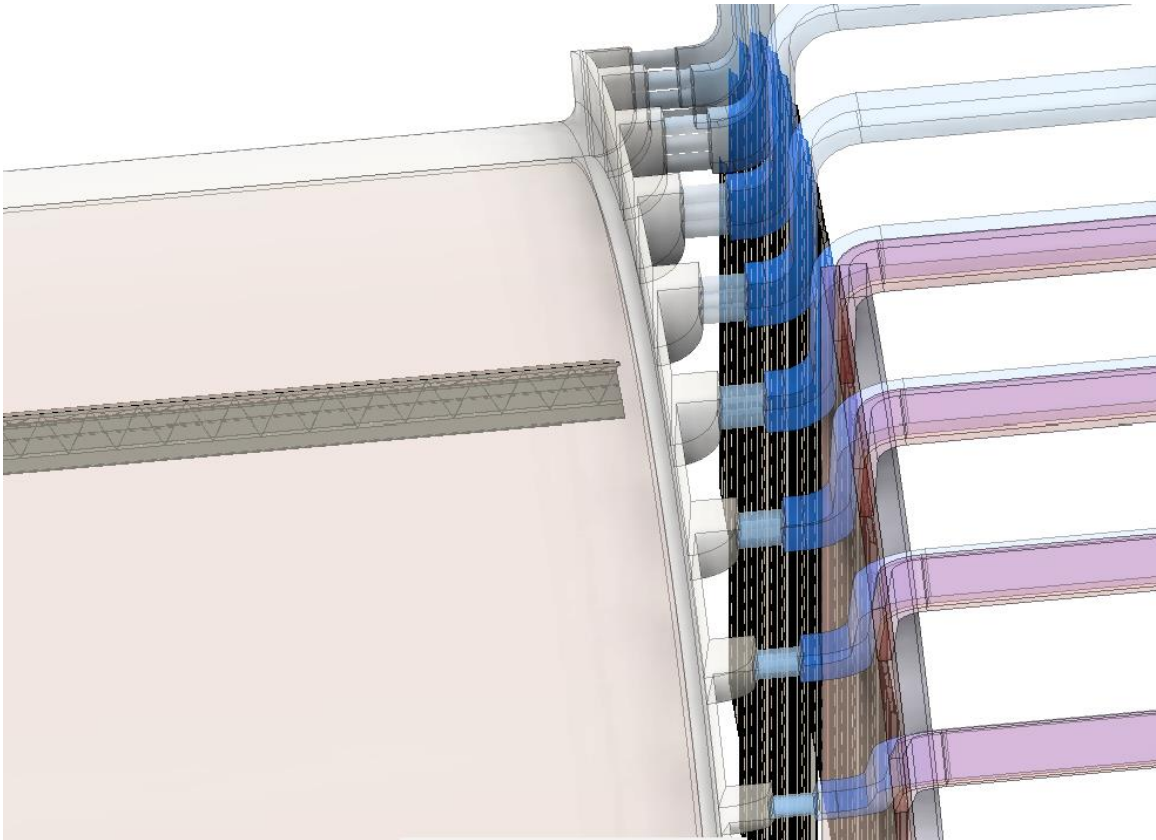
oTOF Half Stave Dimensions	Value V1 R=85cm	Value V2 R=92 cm	Unit
Length	3477.6	3477.6	mm
Width	97.4	97.4	mm
Hight	42.3	42.3	mm
Module overlap	7	7	mm
Number of modules (length)	27	27	
Number of modules (width)	2	2	
Total number of modules	54	54	
Number of sensors	432	432	
Tilt of stave	5	5	°



All Remain
the same

oTOF Envelope – Will need to update this

- oTOF Envelope -> 45mm wide
- oTOF stave height -> 42.3mm
-



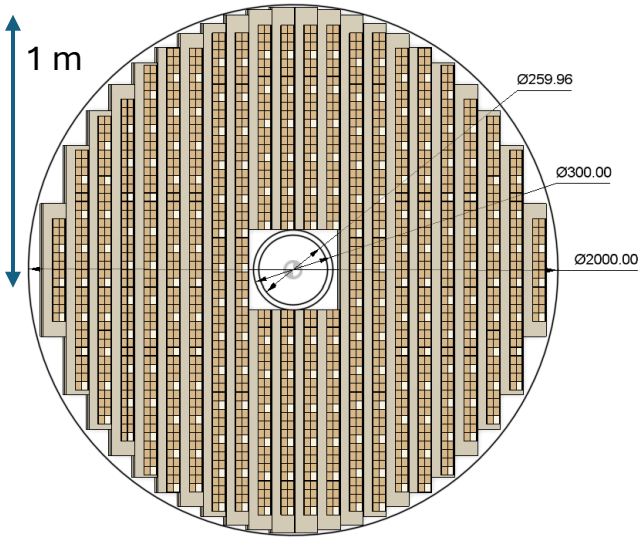
Readout and Services
will need to fit in here!

fTOF

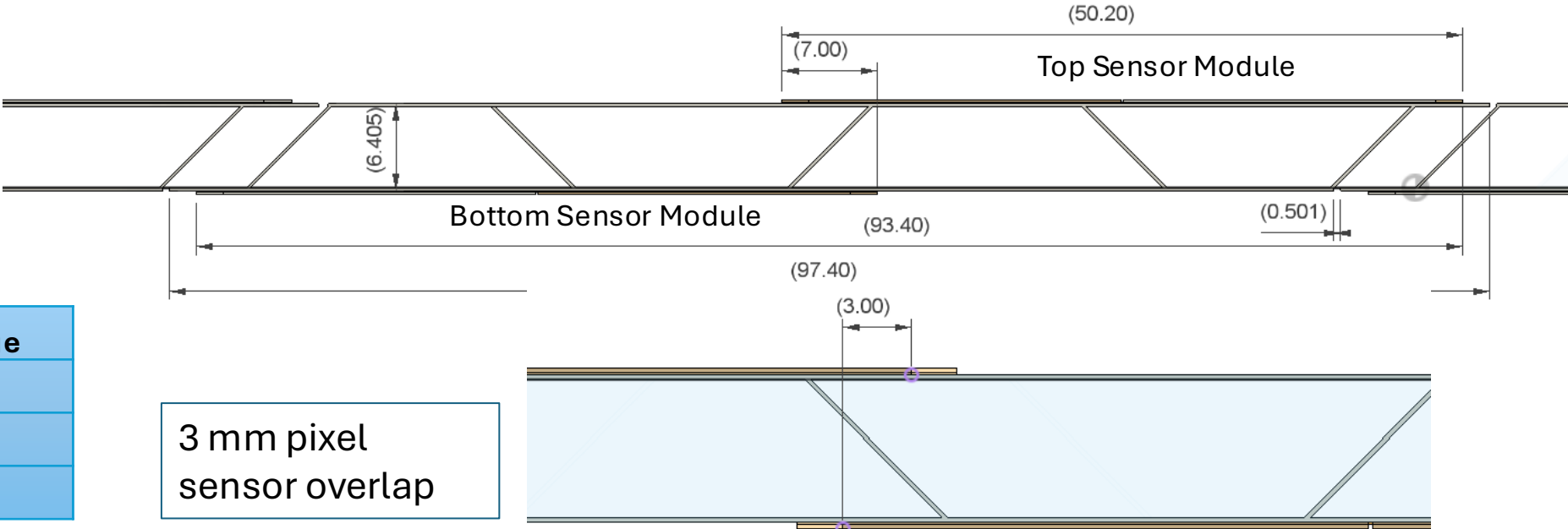
fTOF (One Side)	Value	Unit
Number of panels	26	
Number of unique panels	8	
Number of modules	476	
Total area	2.7	m ²
Coverage	86.3	%

fTOF (A & C Side)	Value
Number of panels	52
Number of unique panels	8
Number of modules	952

3 mm pixel
sensor overlap



2x Forward disks
(A and C side)

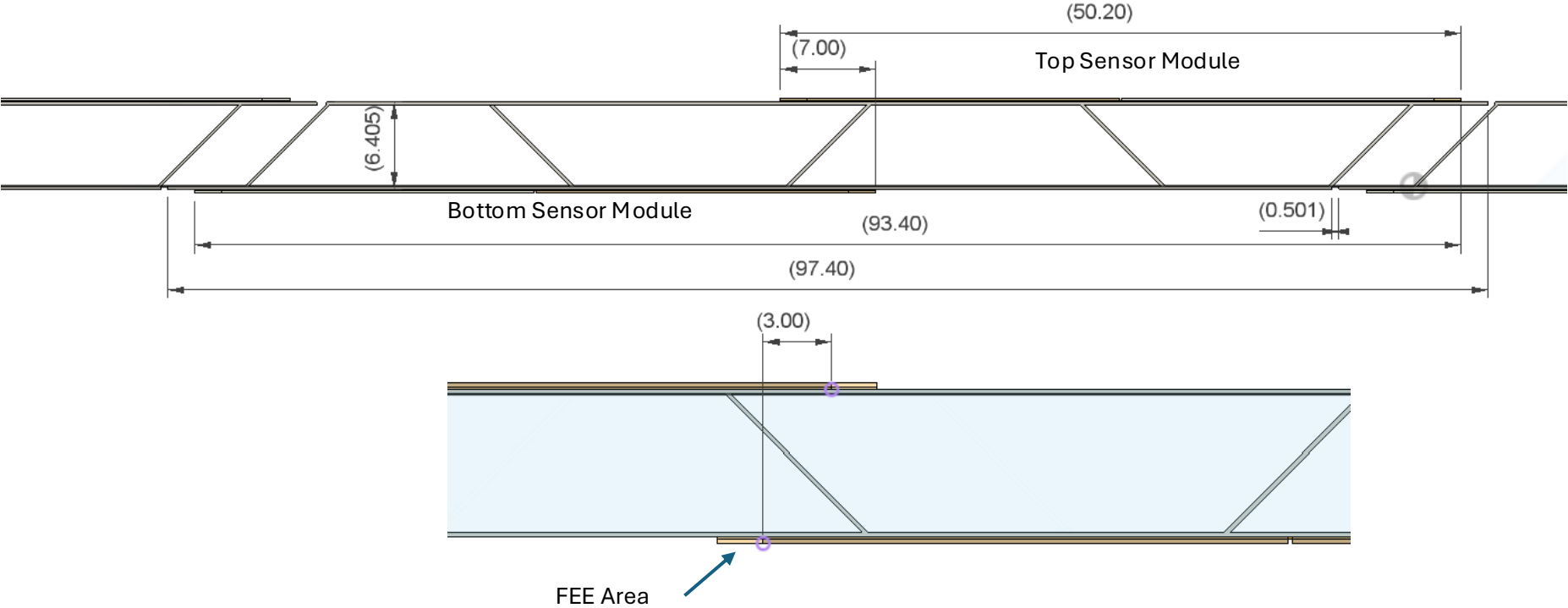
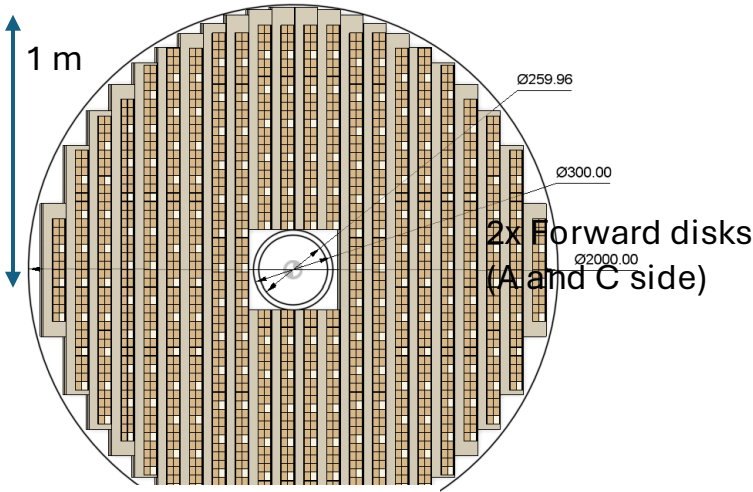


FEE Area

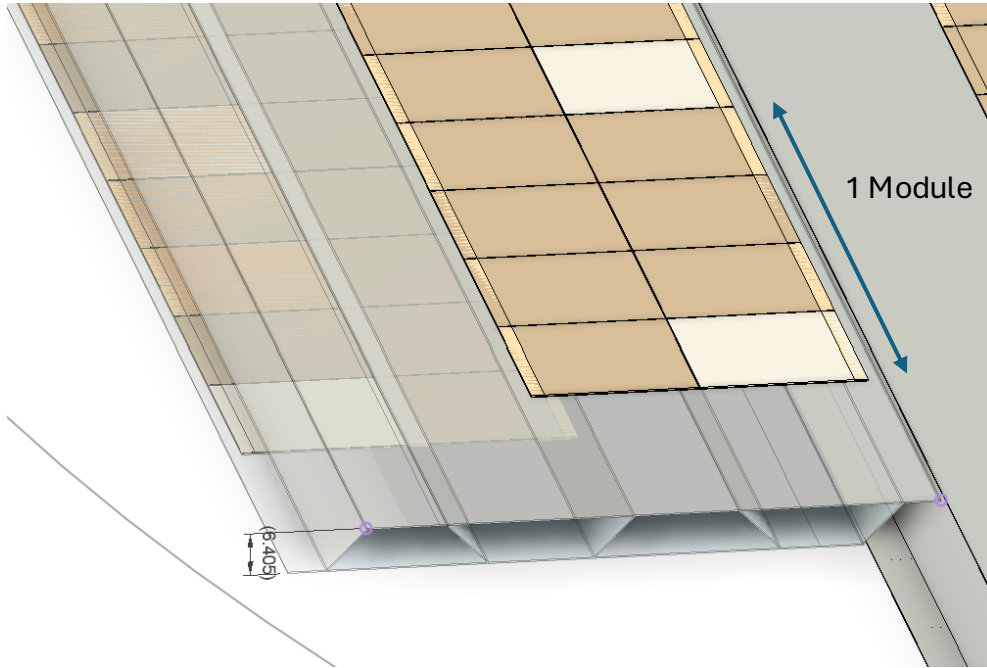
fTOF

fTOF (One Side)	Value	Unit
Number of panels	26	
Number of unique panels	8	
Number of modules	476	
Total area	2.7	m ²
Coverage	22.2	%

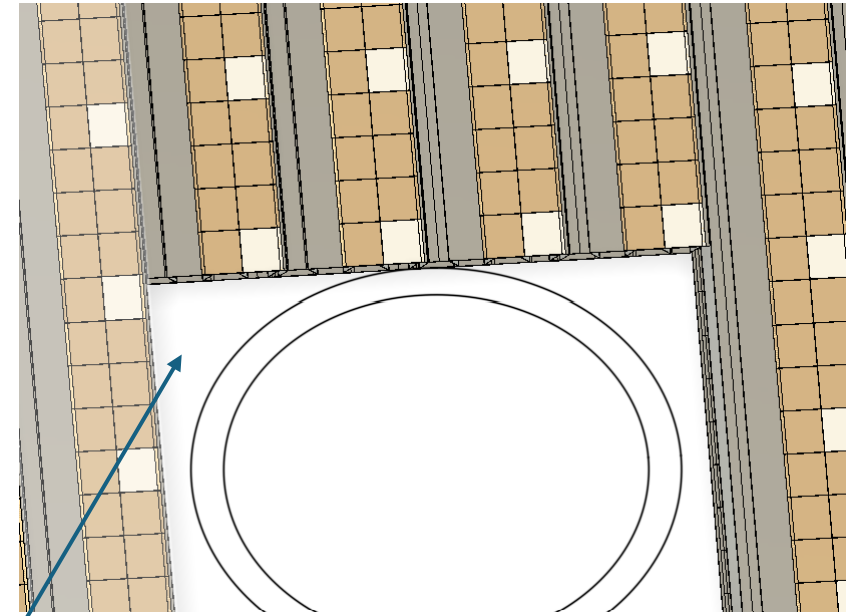
fTOF (A & C Side)	Value
Number of panels	52
Number of unique panels	8
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fTOF



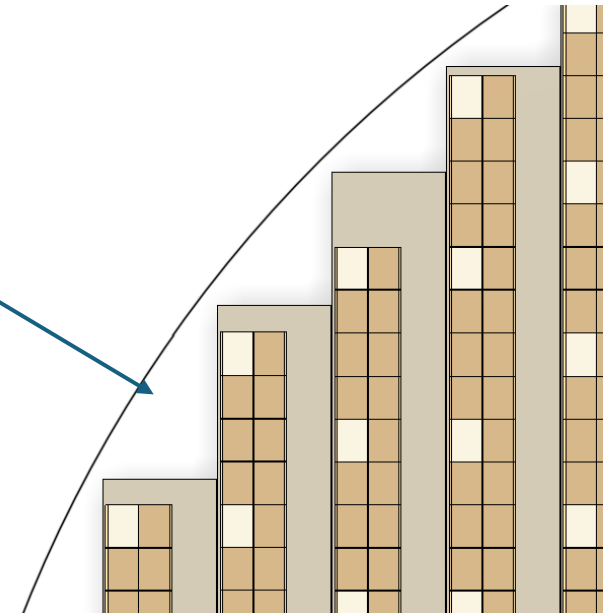
- Modules placed on the front and back of panel
- Same Module used for the iTOF, oTOF and fTOF (2 x 4 sensors)



Some gaps!

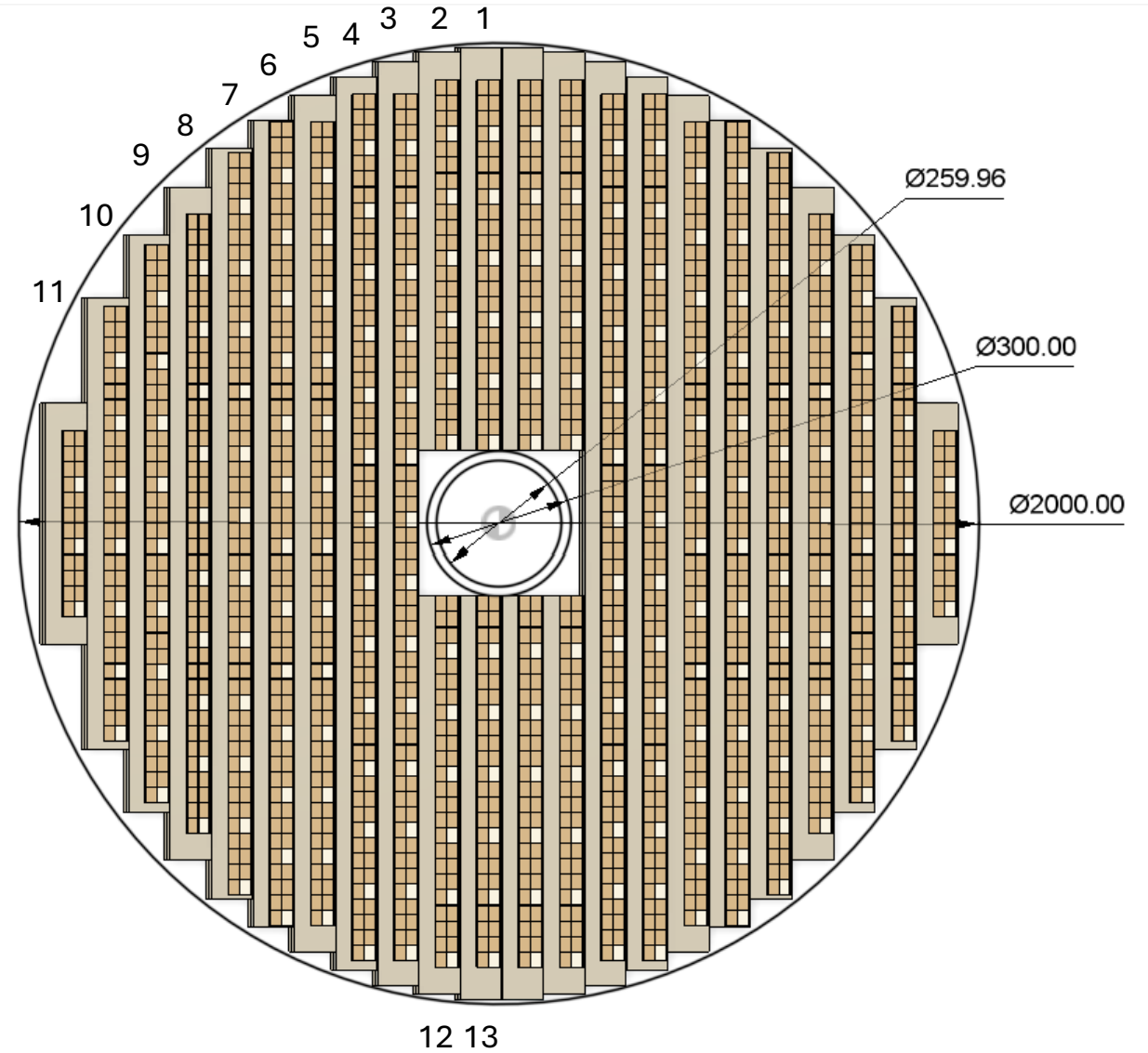
May need to
decrease the
module size for
fTOF

Coverage: 86%

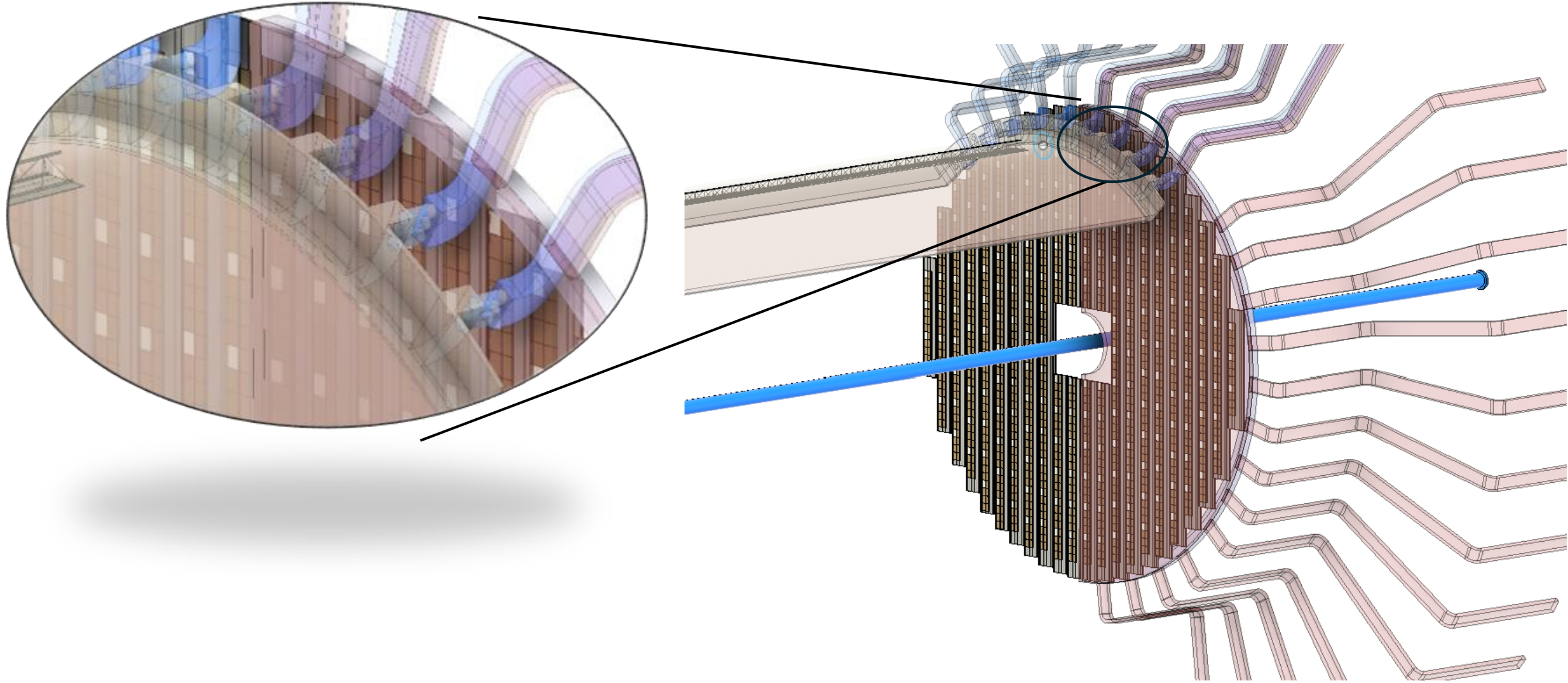


fTOF (One Side)

Panel No.	Length (mm)	Modules
1	772.8	12
2	772.8	12
3	1803.2	28
4	1803.2	28
5	1674.4	26
6	1674.4	26
7	1545.6	24
8	1288	20
9	1159.2	18
10	901.6	14
11	386.4	6
12	772.8	12
13	772.8	12
14	772.8	12
15	772.8	12
16	1803.2	28
17	1803.2	28
18	1674.4	26
19	1674.4	26
20	1545.6	24
21	1288	20
22	1159.2	18
23	901.6	14
24	386.4	6
25	772.8	12
26	772.8	12
Total		476



fTOF Envelope Spacing



Panels can't extend passed the $R:1m$

Thank You

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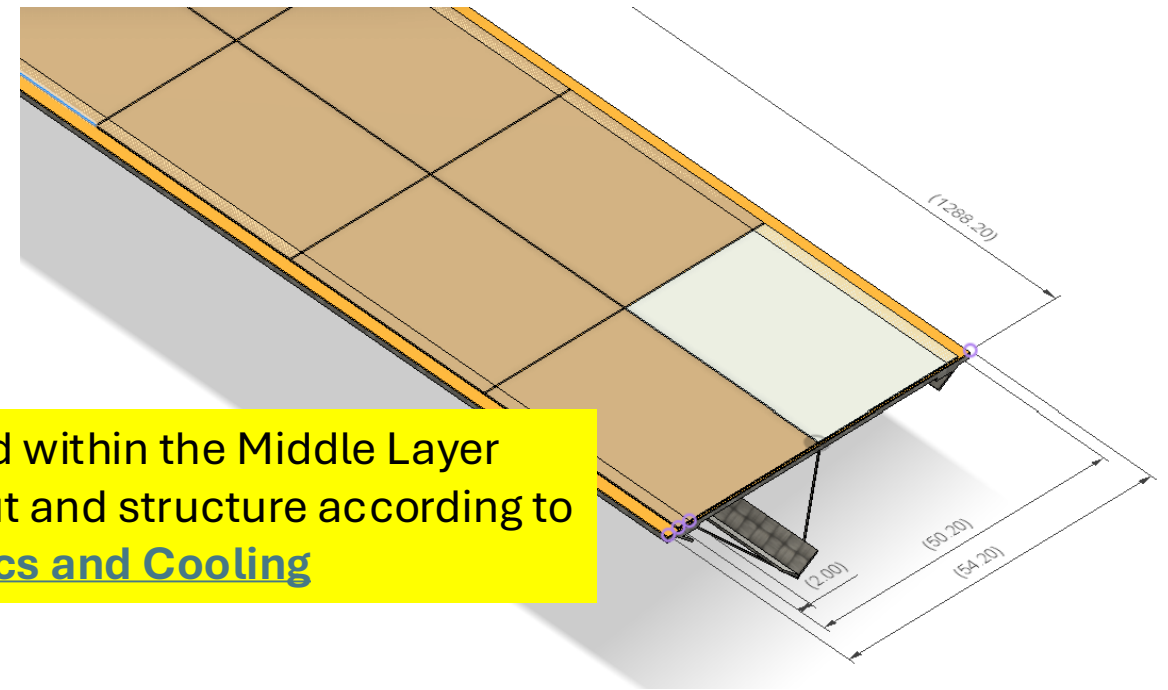
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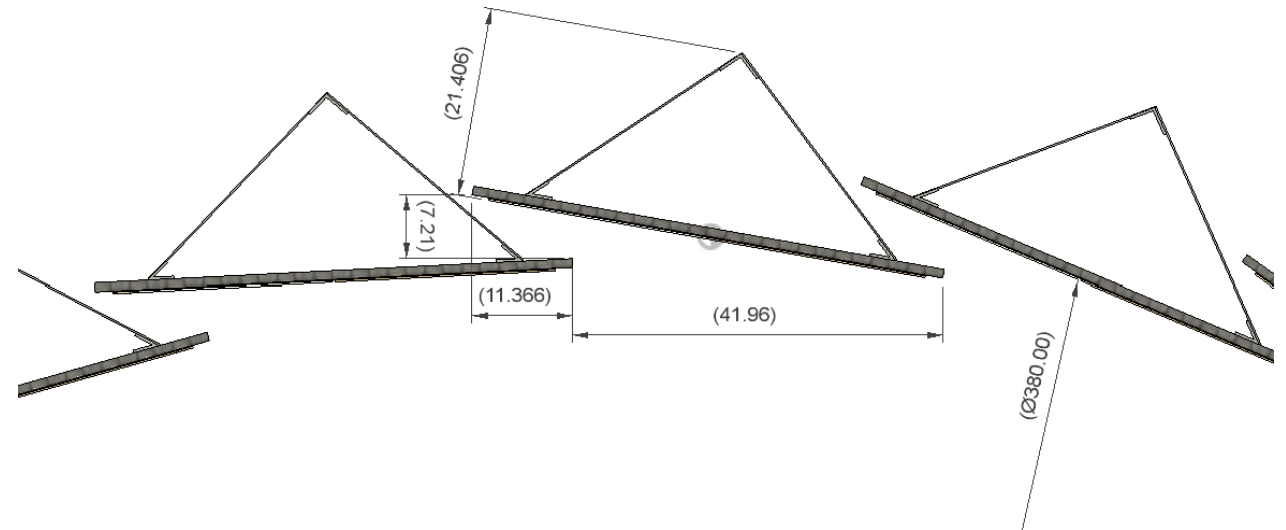
Extra: iTOF Stave Parameters

iTOF Dimensions	Value	Unit
Radius	190	mm
Full Length	1288.2	mm
Circumference	1194	mm
Half stave length	n/a	mm
Surface area	1.538	m2
Number of Staves	24	

iTOF included within the Middle Layer Stack. Layout and structure according to [ML Mechanics and Cooling](#)



iTOF Stave Dimensions	Value	Unit
Length	1288.2	mm
Width	54.2	mm
Hight	21.4	mm
Number of Modules long	10	
Number of modules wide	1	
Total Number of Modules	10	
Number of Sensors	80	
Stave tilt	10	°



Extra: iTOF Staves

